Laboratory Exercise
Syntax Analysis

Objective:

At the end of the exercise, the students should be able to:

- Interpret code structures using parsing algorithms

Requirements:

- Microsoft Word
- IDLE

In Python, some operators are performed before others. It is called the **hierarchy of priorities.**

Priority	Operator
1	** or Exponentiation Operator
2	Unary + and –, where unary operators located next to the right of the power operator bind more strongly. For example, 4 ** **-1** equals 0.25.
3	*, /, //, %
4	Binary + (Addition) and – (Subtraction)

This table enumerates the operators in order from the highest (1) to lowest (4) priorities.

Additionally, Python uses **binding,** which determines the order of computations performed by some operators with equal priority, put side by side in one expression.

Most of Python's operators have left-sided binding, which means that the calculation of the expression is conducted from left to right.

Using the following as an example,

```
print(9 % 6 % 2)
```

There are two possible ways of evaluating this expression:

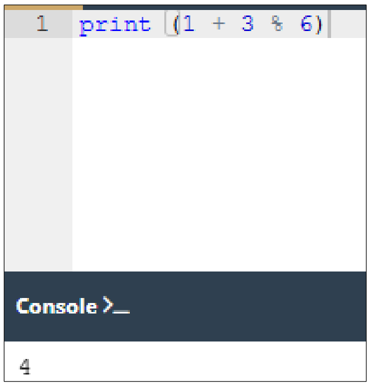
- from left to right: first `9 % 6` gives `3`, and then `3 % 2` gives `1`;
- from right to left: first `6 % 2` gives `0`, and then `9 % 0` causes **a fatal error**.

An exemption is when an exponentiation operator is used as it utilizes right-sided binding.

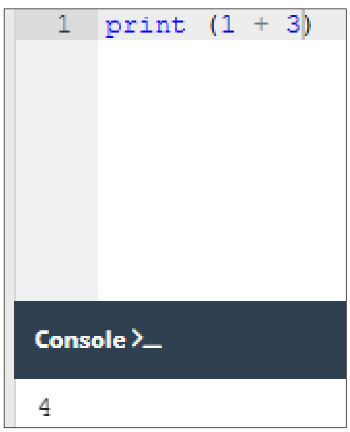
Procedure (60 points):

1. Provide five (5) expressions (when data and operators are connected) that use any two (2) operators from the table on Page 1. For example, 1 + 3 % 6, etc.
2. Launch IDLE (or https://edube.org/sandbox) and create a new Python source file.
3. Run each expression and identify if they are left-sided binding or right-sided binding. Provide screenshots.

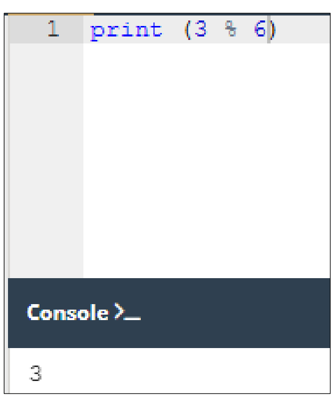
   For example, running `print (1 + 3 % 6)` is right-sided binding as it results in `4`.

   To check, it would be advised to perform each expression first. Performing `print (1 + 3)` results in `4`, which should be the correct answer, but `% 6` still needs to be performed.

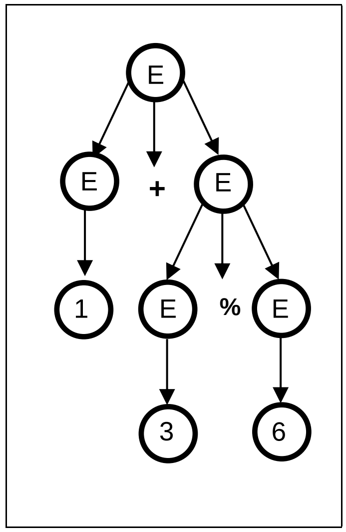
   

   Operating `print(3 % 6)` results in `3.` Then, by adding the remaining `1` value, it results in `4,` which is the correct answer.

It is right-sided binding since it performed the first operator on the right side before the one on the left.

4. Based on the identified binding, create a parse tree for each expression. For instance, `(1 + 3 % 6)` will have this parse tree since it is right-sided binding.



5. Consolidate the screenshots and parse trees in an MS Word file.
6. Call on the instructor to check before submitting a PDF copy on eLMS.

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Criteria	Excellent 4	Good 3	Fair 2	Poor 1	Points
Completeness (x5)	The student provided **all** the requirements.	The student **missed one (1)** requirement.	The student **missed 2-3** requirements.	The student **missed all** of the requirements.	__/20
Screenshots (x5)	The student provided **correct and complete screenshots**.	The student provided **correct but incomplete screenshots**.	The student provided **incorrect but complete screenshots**.	The student provided **incorrect and incomplete screenshots**.	___/20
Parse Trees (x5)	The student provided a **correct and organized parse** tree.	The student provided a **correct but disorganized** parse tree.	The student provided an **organized but incorrect** parse tree.	The student provided an **incorrect and disorganized** parse tree.	___/20
				Total Score	__/60